

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 7

For the tutorial in Week 8 on 14 March, let us discuss the following questions.

Question 1. How many different possible simple graphs of n vertices and m edges are there? Assume that $G = (V, E)$ with $V = \{1, \dots, n\}$.

Question 2.

- (a) Do simple, 3-regular graphs of 5 edges exist?
- (b) How many graphs with 6 vertices and 14 edges (with vertex labels $\{1, \dots, n\}$) are there?

Question 3.

- (a) How many non-isomorphic simple graphs with 3 vertices and 2 edges are there?
- (b) How many non-isomorphic simple graphs with 4 vertices and 2 edges are there?

Question 4. (Ex. 10.2.18.) Show that in a simple graph with at least two vertices, there must be two vertices that have the same degree.

Question 5. How many non-isomorphic simple graphs with 4 vertices and 3 edges are there?

Question 6. (Ex. 10.2.60. and Ex. 10.2.65). The complement of a graph $G = (V, E)$ is the graph $\overline{G} = (V, \overline{E})$ where $\overline{E}$ contains exactly those edges that are not in E .

- (a) Show that if G is a simple graph with n vertices, then the union of G and $\overline{G}$ is K_n .
- (b) If G is a simple graph with 15 edges and $\overline{G}$ has 13 edges, how many vertices does G have?

Question 7. Suppose that G and H are isomorphic simple graphs. Show that their complementary graphs $\overline{G}$ and $\overline{H}$ are also isomorphic.

Question 8. A simple graph G is called self-complementary if G and $\overline{G}$ are isomorphic.

- (a) Show that there are no self-complementary simple graphs with two nor three vertices.
- (b) Find a self-complementary simple graph with four vertices.

Question 9.

- (a) Find a self-complementary simple graph with five vertices.
- (b) For which integers n is C_n self-complementary ?

Question 10. How many subgraphs with at least one vertex does K_3 have?

Answer.

- 1) $C(n(n-1)/2, m)$.
- 2(b) 15.
- 3(a) 1; 3(b) 2.
- 5) 3. Explain why they are not isomorphic.
- 6(b) 8.
- 9(b) 5.
- 10) 17.